DAYANANDA SAGAR UNIVERSITY

SCHOOL OF ENGINEERING

USN No:

II Semester B. Tech Mid Semester Examination
Department of Mechanical Engineering

Course Title: Introduction to Sustainable Engineering **Course Code:** 25EN1120
Duration: 75 Mins.
Date: 13/03/2026
Time: 01:00 PM-02:15 PM
Max Marks: 40

Instructions to Candidates:

- Answer all the four Questions
- Each question carries 10 marks

Q No.		Question Description	Marks	BT Level	COs	POs
1)	a.	Illustrate the role of circular economy concepts in minimizing environmental degradation.	05	L2	CO1	PO1
	b.	Explain the concept of product stewardship. How does it contribute to sustainable development in manufacturing industries?	05	L2	CO1	PO1
2)	a.	Describe the concepts of sustainable sourcing, material efficiency, and embodied energy in engineering products.	04	L2	CO2	PO1
	b.	Define conduction, convection, and radiation mode of heat transfer and discuss its industrial significance.	06	L2	CO2	PO1
3)	a.	List and briefly explain the major core principles of sustainable engineering. → [illegible]	04	L2	CO1	PO1
	b.	Water flows through an inclined pipe of non-uniform cross-section. At the inlet section, the water velocity is 0.6 m/s, the height is $h_1 = 0.8$ m, and the pressure is $P_1 = 1600$ N/m^2. At the outlet section, the water velocity is 0.4 m/s and the height is $h_2 = 0.3$ m. Assuming steady, incompressible flow and neglecting losses, determine the pressure of water at the outlet section P_2.	06	L3	CO2	PO2
4)	a.	A gasoline engine produces 20 hp using 35 kW of energy from burning of fuel. Calculate its thermal efficiency, and how much power is rejected to the ambient?	05	L3	CO2	PO2
	b.	Outline the role of engineers in achieving Sustainable Development Goals (SDGs).	05	L2	CO1	PO1

DAYANANDA SAGAR UNIVERSITY

SCHOOL OF ENGINEERING

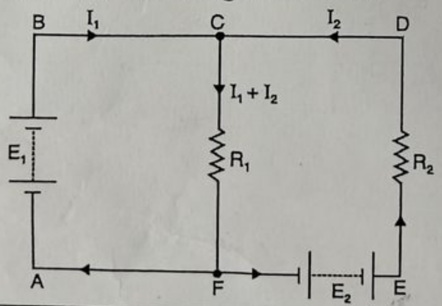
USN No: ENG25CS0453

B. Tech II Semester End Examinations – May 2026

Course Title: Introduction to Electrical and Electronics Engineering
Course Code: 25EN1109
Duration: 02 Hours 30 Minutes
Time: 01:00 PM to 03:30 PM

Date: 18-05-2026
Max Marks: 80

Note:
1. **Answer all Five Full Questions**
2. **Each Full Question carries 16 Marks**
3. **Draw neat sketches wherever necessary**
4. **Missing Data may be suitably assumed**

Q. No.	Question		Marks	BTL	COs
1.a.	Draw the circuit diagram for the parallel connection of two resistors and write the expression for equivalent resistance. Describe the current division rule of parallel resistors.		06	L2	CO1
1.b.	A battery has an e.m.f. of 12.8 V and supplies a current of 3.2 A. Determine: (i) resistance of the circuit (ii) power output from the battery		04	L3	CO2
1.c.	Give the statement of Kirchhoff's Current Law (KCL) and write the KVL equation for the circuit given below. 		06	L3	CO1
2.a.	Describe the following: (i) Magnetic Flux Density (ii) Faraday's 1st law of electromagnetism (iii) Mutually induced EMF (iv) Non-conventional sources of energy		04	L1	CO2

(P.T.O.)